Beamforming Optimization for Cell-Free ISAC Systems: Mathematical Derivation and Solution Methods

Research Stage Report

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Abstract

This paper establishes a mathematical framework for joint beamforming optimization in Cell-Free Integrated Sensing and Communication (ISAC) systems. Considering communication quality-of-service (QoS) constraints, sensing detection probability constraints, tracking accuracy constraints (PCRB), and access point (AP) selection constraints, we derive the semidefinite programming (SDP) relaxation formulation of the robust optimization problem under imperfect channel state information (CSI). Through the S-Procedure, infinite-dimensional worst-case constraints are transformed into finite-dimensional linear matrix inequalities (LMI). The convexity of sensing constraints under the single-angle estimation assumption is rigorously proved, and rank-one recovery algorithms along with closed-form solution schemes are provided. This document serves as the theoretical foundation for subsequent algorithm implementation and paper writing.

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1 System Modeling and Problem Formulation (Plate I: Physical Modeling)

1.1 Optimization Problem (Standard Form)

Consider a Cell-Free ISAC system with M distributed APs, K communication users, and P sensing targets. Each AP is equipped with N_t transmit antennas. The optimization objective is to minimize the total transmit power while satisfying communication QoS, sensing detection, tracking accuracy, and AP selection constraints.

$$\min_{\substack{\{\mathbf{w}_{m,k}\},\{\mathbf{Z}_m\},\\ \{b_{mp}\}}} \quad \sum_{m=1}^{M}\left(\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m)\right) \tag{P1}$$

$$\text{s.t.} \quad \mathrm{SINR}_k^{\mathrm{wc}} \geq \gamma_k, \quad \forall k \in \mathcal{K} \tag{1}$$

$$\mathrm{SINR}_{S,p} \geq \gamma_S^{\mathrm{PoD}}, \quad \forall p \in \mathcal{P} \tag{2}$$

$$\mathrm{tr}(\mathbf{J}_p^{\mathrm{data}}) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P} \tag{3}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{4}$$

$$\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m) \leq P_{\max}, \quad \forall m \in \mathcal{M} \tag{5}$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \tag{6}$$

$$b_{mp} \in \{0,1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \tag{7}$$

where $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ denotes the communication beamforming vector from AP m to user k, $\mathbf{Z}_m \in \mathbb{C}^{N_t \times N_t}$ represents the sensing covariance matrix at AP m, and $b_{mp} \in \{0,1\}$ is the binary indicator variable for whether AP m serves target p. The parameters $P_{\max}$, γ_k, γ_S^{PoD}, $\Gamma_{\mathrm{Track},p}$, and N_{req} denote the per-AP maximum transmit power, the SINR threshold for user k, the sensing detection probability threshold, the tracking accuracy threshold (FIM trace constraint) for target p, and the required number of serving APs per target, respectively.

1.2 Notation Table

Table 1 summarizes the main notation used throughout this paper.

Remark 1. *Specific numerical parameters (e.g., $M = 16$, $N_t = 4$, $P_{\max} = 30$ W) will be provided in the Simulation Setup section. The theoretical derivations herein maintain general symbolic notation.*

2 System Model and Imperfect CSI

As the second step of physical modeling, this section characterizes the transmission of communication and sensing signals under imperfect channel state information.

Table 1: Main Notation Table

Symbol	Dimension	Physical Meaning
M	Scalar	Total number of APs
N_t	Scalar	Number of transmit antennas per AP
K	Scalar	Total number of communication users
P	Scalar	Total number of sensing targets
$P_{\max}$	Scalar	Maximum transmit power per AP
γ_k	Scalar	SINR threshold for user k
γ_S^{PoD}	Scalar	Sensing detection probability threshold
$\Gamma_{\mathrm{Track},p}$	Scalar	Tracking accuracy threshold for target p
σ_c^2	Scalar	Communication receiver noise variance
σ_s^2	Scalar	Sensing receiver noise variance
ϵ_h	Scalar	Relative error bound for communication channel estimation
ϵ_g	Scalar	Relative error bound for sensing channel estimation
N_{req}	Scalar	Required number of serving APs per target
$\mathbf{h}_{m,k}$	$\mathbb{C}^{N_t}$	Communication channel from AP m to user k
$\mathbf{g}_{m,p}$	$\mathbb{C}^{N_t}$	Sensing channel from AP m to target p
$\mathbf{w}_{m,k}$	$\mathbb{C}^{N_t}$	Communication beam from AP m to user k
$\mathbf{Z}_m$	$\mathbb{C}^{N_t\times N_t}$	Sensing covariance matrix at AP m
b_{mp}	$\{0,1\}$	Whether AP m serves target p

2.1 Communication Signal Model

The received signal at user k can be expressed as:

$$y_k = \underbrace{\sum_{m=1}^{M} \mathbf{h}_{m,k}^H \mathbf{w}_{m,k} s_k}_{\text{desired signal}} + \underbrace{\sum_{j\neq k}\sum_{m=1}^{M} \mathbf{h}_{m,k}^H \mathbf{w}_{m,j} s_j}_{\text{multi-user interference}} + n_k \tag{8}$$

where $n_k \sim \mathcal{CN}(0,\sigma_c^2)$ denotes the receiver noise.

To facilitate analysis, we introduce stacked vector forms. Define the global communication channel $\mathbf{h}_k \in \mathbb{C}^{MN_t}$ and global communication beam $\mathbf{w}_k \in \mathbb{C}^{MN_t}$ as:

$$\mathbf{h}_k = [\mathbf{h}_{1,k}^T, \mathbf{h}_{2,k}^T, \ldots, \mathbf{h}_{M,k}^T]^T, \quad \mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \ldots, \mathbf{w}_{M,k}^T]^T \tag{9}$$

The received signal can then be compactly written as:

$$y_k = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{j\neq k} \mathbf{h}_k^H \mathbf{w}_j s_j + n_k \tag{10}$$

2.2 Sensing Signal Model

For monostatic radar sensing, the reflected signal from target p is:

$$y_{S,p} = \sum_{m=1}^{M} \mathbf{g}_{m,p}^H \mathbf{Z}_m \mathbf{g}_{m,p} + n_{S,p} \tag{11}$$

where $n_{S,p} \sim \mathcal{CN}(0,\sigma_s^2)$ denotes the sensing receiver noise.

2.3 Imperfect CSI Model

In practical systems, channel estimation suffers from errors. This paper adopts the bounded error model:

Communication channel:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta\mathbf{h}_k, \quad \|\Delta\mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2 \tag{12}$$

Sensing channel:

$$\mathbf{g}_p = \hat{\mathbf{g}}_p + \Delta\mathbf{g}_p, \quad \|\Delta\mathbf{g}_p\|_2 \leq \epsilon_g \|\hat{\mathbf{g}}_p\|_2 \tag{13}$$

where $\hat{\mathbf{h}}_k$ and $\hat{\mathbf{g}}_p$ are the estimated channels, $\Delta\mathbf{h}_k$ and $\Delta\mathbf{g}_p$ are the estimation errors, and ϵ_h and ϵ_g are the relative error bounds.

3 Communication Constraint Derivation

The third step of physical modeling: starting from the signal model, we derive the original non-convex expressions of worst-case SINR, sensing SINR, and PCRB. These constraints will be convexified in Section III.

3.1 SINR Definition

The signal-to-interference-plus-noise ratio (SINR) for communication user k is defined as:

$$\mathrm{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j\neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \sigma_c^2} \tag{1}$$

3.2 Worst-Case SINR Analysis

Under imperfect CSI, we need to guarantee that the worst-case SINR meets the threshold requirement. Considering the worst cases for both the numerator and interference:

Worst-case numerator: When the error direction opposes the beam, the effective channel gain is minimized:

$$|\mathbf{h}_k^H \mathbf{w}_k|^2 \approx |\hat{\mathbf{h}}_k^H \mathbf{w}_k|^2 (1 - \epsilon_h)^2 \tag{14}$$

Worst-case interference: When the error direction enhances interference:

$$|\mathbf{h}_k^H \mathbf{w}_j|^2 \approx |\hat{\mathbf{h}}_k^H \mathbf{w}_j|^2 (1 + \epsilon_h)^2 \tag{15}$$

Therefore, the worst-case SINR can be approximated as:

$$\mathrm{SINR}_k^{\mathrm{wc}} \approx \mathrm{SINR}_k^{\mathrm{nom}} \cdot \frac{(1 - \epsilon_h)^2}{(1 + \epsilon_h)^2} \tag{2}$$

3.3 Robustness Factor and Threshold

Define the communication robustness factor:

$$\eta_h = \left(\frac{1-\epsilon_h}{1+\epsilon_h}\right)^2 \tag{3}$$

For $\epsilon_h = 0.10$ (10% error), $\eta_h \approx 0.669$ (-1.75 dB). This implies that CSI errors cause an effective SINR degradation of approximately 1.75 dB.

To ensure the worst-case SINR meets the threshold, the nominal SINR requirement must be raised:

$$\gamma_k^{\text{robust}} = \gamma_k \cdot \frac{(1+\epsilon_h)^2}{(1-\epsilon_h)^2} = \frac{\gamma_k}{\eta_h} \tag{4}$$

For $\gamma_k = 1$ (0 dB) and $\epsilon_h = 0.10$: $\gamma_k^{\text{robust}} \approx 1.493$ (1.74 dB).

4 Sensing Constraint Derivation

4.1 Sensing SINR

The sensing SINR for target p is:

$$\text{SINR}_{S,p} = \frac{|\mathbf{g}_p^H \mathbf{z}_p|^2}{\sigma_s^2 \|\mathbf{z}_p\|_2^2} \tag{5}$$

By the Cauchy-Schwarz inequality, the optimal sensing beam is the matched filter:

$$\mathbf{z}_p^* = \sqrt{P_{S,p}} \frac{\mathbf{g}_p}{\|\mathbf{g}_p\|_2} \tag{6}$$

yielding the optimal SINR:

$$\text{SINR}_{S,p}^* = \frac{P_{S,p}\|\mathbf{g}_p\|_2^2}{\sigma_s^2} \tag{7}$$

To meet the detection threshold γ_S^{PoD}, the minimum sensing power is:

$$P_{S,p}^{\min} = \frac{\gamma_S^{\text{PoD}} \sigma_s^2}{\|\mathbf{g}_p\|_2^2} \tag{8}$$

4.2 Robust Sensing Constraint

Similarly, define the sensing robustness factor:

$$\eta_g = \left(\frac{1-\epsilon_g}{1+\epsilon_g}\right)^2 \tag{9}$$

For $\epsilon_g = 0.15$ (15% error), $\eta_g \approx 0.546$ (-2.63 dB).
The robust sensing threshold:

$$(\gamma_S^{\text{PoD}})^{\text{robust}} = \gamma_S^{\text{PoD}} \cdot \frac{(1+\epsilon_g)^2}{(1-\epsilon_g)^2} = \frac{\gamma_S^{\text{PoD}}}{\eta_g} \tag{10}$$

4.3 PCRB Constraint and Fisher Information Matrix

The tracking accuracy constraint is characterized through the trace of the Fisher Information Matrix (FIM). The data-dependent FIM is:

$$\mathbf{J}_p^{\text{data}} = \frac{2}{\sigma_s^2}\text{Re}\left\{\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \cdot \mathbf{R}_X \cdot \nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p\right\} \tag{11}$$

where $\mathbf{R}_X = \sum_k \mathbf{w}_k\mathbf{w}_k^H + \sum_p \mathbf{z}_p\mathbf{z}_p^H = \sum_k \mathbf{W}_k + \mathbf{Z}$ is the transmit covariance matrix and $\boldsymbol{\theta}_p$ denotes the target state parameters.

Assumption 1 (Sensing Parameter Estimation Model)**.** *This paper considers target state parameters $\boldsymbol{\theta}_p = [\theta_p]$ (single-angle estimation). Under this setting, the gradient matrix $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p$ is determined by the target prediction state within the current time slot and can be treated as a known constant matrix. Consequently, $\mathbf{J}_p^{data}$ is an **affine function** of $\mathbf{R}_X$, and its trace constraint can be precisely formulated as:*

$$tr(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{Track,p} \tag{16}$$

where $\mathbf{F}_p = \frac{2}{\sigma_s^2}Re\left\{\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\right\}$ is a known constant positive semidefinite matrix.

Remark 2. *Since this work focuses on direction-of-arrival (DOA) estimation for targets, the relative delay and Doppler shift between the probing signal and the target are regarded as slowly varying parameters within a short observation frame. Under this condition, the FIM degenerates into a constant matrix $\mathbf{F}_p$ that depends solely on the angle gradient, thereby guaranteeing the convex affine property of the constraint.*

Proof of FIM Linearity: Let $\mathbf{G}_p = \nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \in \mathbb{C}^{D\times MN_t}$, then:

$$\mathbf{J}_p^{\text{data}} = \frac{2}{\sigma_s^2}\text{Re}\{\mathbf{G}_p\mathbf{R}_X\mathbf{G}_p^H\} \tag{17}$$

Expanding the (a, b)-th element:

$$(\mathbf{J}_p^{\text{data}})_{ab} = \frac{2}{\sigma_s^2}\text{Re}\left\{\sum_{i,j}(\mathbf{G}_p)_{ai}(\mathbf{R}_X)_{ij}(\mathbf{G}_p^H)_{jb}\right\} \tag{18}$$

This is a linear combination of the elements of $\mathbf{R}_X$. Therefore, the trace constraint:

$$\text{tr}(\mathbf{J}_p^{\text{data}}) = \text{tr}(\mathbf{F}_p\mathbf{R}_X) \tag{12}$$

where:

$$\mathbf{F}_p = \frac{2}{\sigma_s^2}\text{Re}\left\{\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\right\} \tag{13}$$

is a constant positive semidefinite matrix independent of the optimization variables. □

5 SDP Relaxation and Global Variable Reformulation (Plate II: Mathematical Reformulation)

The physical modeling in Section II yields a highly non-convex problem (P1). This section presents a seven-step convexification chain that transforms (P1) into a standard convex SDP (P3). The equivalence proof of each step is given explicitly in §III.2.

5.0.1 Identification of Non-convex Sources

To make the subsequent convexification derivations amenable to rigorous review, this subsection pre-identifies all non-convex sources in the original problem and provides the mathematical justification for each. The original problem (P1) carries constraint labels (5a)–(5h); see §**??**.

Table 2: Identification of non-convex sources in problem (P1)

ID	Non-convex source	Affected constraints	Convexity violation type
NC1	SINR fractional structure	(5b), (5c)	Convex set not closed under multiplic
NC2	Semi-infinite robust constraints	(5b), (5c)	Infinite constraints violate convexity
NC3	Binary combinatorial constraints	(5h)	Discrete set $\{0, 1\}$ is non-convex
NC4	Implicit rank-1 constraints	(5a)–(5f)	Rank-1 set is non-convex
NC5	Beam-channel bilinear coupling	(5b), (5d)	Bilinear functions are non-convex
NC6	Matrix inverse constraints	(5d)	Inverse map does not preserve convex

Mathematical criterion for convexity: A set $\mathcal{C}$ is convex iff $\forall \mathbf{x}, \mathbf{y} \in \mathcal{C}, \forall \theta \in [0, 1]$: $\theta\mathbf{x} + (1 - \theta)\mathbf{y} \in \mathcal{C}$. Using this criterion:

- $\|\mathbf{w}\|_2^2 \leq \alpha$: convex (second-order cone)
- $\mathrm{tr}(\mathbf{H}_k\mathbf{W}_k) \geq \beta$ with $\mathbf{W}_k \succeq \mathbf{0}$: convex (PSD cone + linear function)
- $\mathrm{rank}(\mathbf{W}_k) = 1$: **non-convex** (rank-1 set is not affine)
- $\frac{x^2}{y} \leq \alpha, y > 0$: **non-convex** (fractional structure does not preserve convexity)
- $b \in \{0, 1\}$: **non-convex** (discrete point set)

5.0.2 Seven-Step Convexification Chain

Targeting the six non-convex sources identified above, this subsection presents a numbered seven-step convexification chain. Each step specifies: which non-convex source it eliminates, the pre- and post-transformation mathematical forms, the transformation property (strictly equivalent / tight relaxation / conservative approximation), and a proposition with proof.

Step 1: Global Variable Lifting (eliminate NC1, NC5) Transformation type: **Strictly equivalent**.

Lift the vector beam $\mathbf{w}_k$ into a covariance matrix:

$$\mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H \in \mathbb{C}^{MN_t \times MN_t} \tag{15}$$

$$\mathbf{Z} = \sum_p \mathbf{z}_p\mathbf{z}_p^H \in \mathbb{C}^{MN_t \times MN_t} \tag{16}$$

Proposition 1 (Lifting equivalence)**.** *When $\mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H$, we have tr$(\mathbf{H}_k\mathbf{W}_k) = |\mathbf{h}_k^H\mathbf{w}_k|^2$, and the lifting $\mathbf{w}_k \mapsto \mathbf{W}_k$ is a one-to-one map.*

Convexity impact: The objective $\sum_k \|\mathbf{w}_k\|^2 = \sum_k \mathrm{tr}(\mathbf{W}_k)$ becomes a linear function of $\mathbf{W}_k$, so the **objective becomes convex**.

Step 2: SDR Relaxation (eliminate NC4) **Transformation type**: **Tight relaxation**. Strictly equivalent for $K \leq 2$; lower bound for $K > 2$.

The rank-1 constraint $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$ is equivalent to

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \mathrm{rank}(\mathbf{W}_k) = 1 \tag{19}$$

SDR drops the rank-1 constraint, requiring only

$$\mathbf{W}_k \succeq \mathbf{0} \tag{S2.1}$$

The feasible set enlarges from $\mathcal{F}_{\text{rank-1}} = \{\mathbf{W}_k \succeq \mathbf{0} : \mathrm{rank}(\mathbf{W}_k) = 1\}$ to $\mathcal{F}_{\text{SDR}} = \{\mathbf{W}_k \succeq \mathbf{0}\}$ (the convex PSD cone).

Proposition 2 (SDR tightness conditions)**.** *SDR is equivalent to the original problem iff the optimal solution satisfies* $\mathrm{rank}(\mathbf{W}_k^*) = 1$*.*

1. *SDR is tight for $K \leq 2$ (guaranteed by Luo's results on MISO multicast).*
2. *SDR is approximately tight in the high-SNR regime.*
3. *For general $K > 2$: Gaussian randomization ($\xi_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$) extracts a feasible rank-1 solution with $O(1/L)$ performance loss.*

Equivalence/lower/upper bound: strictly equivalent when tight; otherwise a **lower bound** on the objective (enlarged feasible set $\Rightarrow$ optimal value $\leq$ original).

Step 3: S-Procedure for Robust SINR (eliminate NC2) **Transformation type**: **Conservative approximation**. Trades a 1.75 dB power margin for a closed-form LMI.

The worst-case SINR constraint is a semi-infinite optimization:

$$\min_{\|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2} \frac{|(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{W}_k (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)|}{\sum_{j \neq k} |(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{W}_j (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)| + \sigma_c^2} \geq \gamma_k \tag{20}$$

Lemma 1 (Cauchy-Schwarz worst case)**.** *For any* $\mathbf{x}, \mathbf{y}$ *and* $\Delta \mathbf{y}$ *with* $\|\Delta \mathbf{y}\| \leq \epsilon \|\mathbf{y}\|$*:*

$$\min_{\|\Delta \mathbf{y}\| \leq \epsilon \|\mathbf{y}\|} |\mathbf{x}^H (\mathbf{y} + \Delta \mathbf{y})|^2 = |\mathbf{x}^H \mathbf{y}|^2 (1 - \epsilon)^2 \tag{21}$$

Proof: By $|\mathbf{x}^H \Delta \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\Delta \mathbf{y}\| \leq \|\mathbf{x}\| \cdot \epsilon \|\mathbf{y}\|$*, with equality at* $\Delta \mathbf{y} = -\epsilon \frac{\|\mathbf{y}\|}{\|\mathbf{x}\|} \mathbf{x}$*.* $\square$

Applying Lemma 1 (lower-bounding the numerator, upper-bounding the denominator) and cross-multiplying, the worst-case robust constraint becomes a linear constraint on a robust threshold γ_k^{robust} (see Eq. (31)).

Equivalence/lower/upper bound: conservative upper bound. The true worst-case SINR is higher than this approximation $\Rightarrow$ the solved feasible set is slightly larger than the true robust feasible set, but robustness is guaranteed under true channel realizations. The cost is approximately 1.75 dB power margin (for $\epsilon_h = 0.10$, $\eta_h \approx 0.669$).

Step 4: SINR Fraction Linearization (eliminate NC1, communication part) **Transformation type**: **Strictly equivalent** (provided the denominator > 0).

After Step 3's robust approximation, the SINR is still fractional. Cross-multiplication yields a linear matrix inequality:

$$\mathrm{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k^{\mathrm{robust}} \sum_{j \neq k} \mathrm{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) \geq \gamma_k^{\mathrm{robust}} \sigma_c^2 \tag{S4.1}$$

where $\hat{\mathbf{H}}_k = \hat{\mathbf{h}}_k \hat{\mathbf{h}}_k^H$.

Proposition 3 (Fraction linearization equivalence)**.** *When the denominator* $\sum_{j \neq k} |\hat{\mathbf{h}}_k^H \mathbf{W}_j \hat{\mathbf{h}}_k| + \sigma_c^2 > 0$*,*

$$\frac{A}{B} \geq \gamma \iff A \geq \gamma B \tag{22}$$

Convexity impact: Eq. (31) (equivalent to (S4.1) after S-Procedure lifting) becomes an LMI in $\mathbf{W}_k$, so the **communication constraint becomes convex**.

Step 5a: PCRB Affine Expansion (eliminate NC6) **Transformation type**: **Strictly equivalent** (under Assumption 1).

The Fisher information matrix is an affine function of $\mathbf{R}_X$. Define

$$\mathbf{F}_p = \frac{2}{\sigma_s^2} \mathrm{Re}\Big\{ \nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p \cdot \nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p^H \Big\} \in \mathbb{S}_+^{D \times D} \tag{13}$$

The PCRB constraint becomes

$$\mathrm{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p} \tag{S5.1}$$

Proposition 4 (PCRB linearization equivalence)**.** *Under Assumption 1 ($\nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p$ is known within the current time slot), tr($\mathbf{J}_p^{data}$) = tr($\mathbf{F}_p \mathbf{R}_X$).*

Convexity impact: Eq. (S5.1) is a linear constraint in $\mathbf{R}_X$, so the **sensing PCRB constraint becomes convex**.

Step 5b: Sensing SINR Linearization **Transformation type**: **Strictly equivalent** (rank-1 optimal solution is matched filtering).

In the sensing SINR constraint $\mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2$, the matrix $\mathbf{g}_p \mathbf{g}_p^H$ is a known constant PSD matrix and $\mathrm{tr}(\cdot)$ is linear in $\mathbf{Z}$.

Proposition 5 (Sensing SINR linearization equivalence)**.** *When $\mathbf{Z} = \mathbf{z}\mathbf{z}^H$ (rank-1), the optimal sensing beam is $\mathbf{z}^* = \sqrt{P_S}\mathbf{g}_p / \|\mathbf{g}_p\|$ (matched filtering), and the constraint holds with equality.*

Step 6: Power Constraint Is Already Convex The power constraint $\sum_k \mathrm{tr}(\mathbf{W}_k) + \mathrm{tr}(\mathbf{Z}_m) \leq P_{\max}$ is linear in $\mathbf{W}_k, \mathbf{Z}_m$, **already convex**; no transformation needed.

Step 7: AP Selection Two-Step Decomposition (handle NC3) **Transformation type**: **Heuristic lower bound**. The outer discrete AP assignment is NP-hard; the inner continuous problem is a convex SDP for any fixed AP set.

This work adopts a two-step decomposition:

1. **Outer (discrete)**: sort APs by large-scale fading $\mathrm{PL}(d_{m,p})$ and select the top-N_{req} APs to fix $\mathcal{M}^{\mathrm{all}}$.

2. **Inner (convex)**: solve the convex SDP (P3) on the fixed AP set.

Remark 3 (Engineering trade-off for AP selection)**.** *The AP selection problem is NP-hard (a variant of the K-medoid problem). This work uses an outer heuristic plus inner convex SDP to obtain a polynomial-complexity, engineering-acceptable sub-optimal solution. The complexity lower bound is established in the companion document [**?**] §6.*

5.0.3 Convexification Chain Summary

The following table summarizes the transformation type and performance impact of each step:

Table 3: Summary of the seven-step convexification chain

Step	Transformation	Eliminates	Type
1	Global variable lifting $\mathbf{w}\mathbf{w}^H \to \mathbf{W}$	NC1, NC5	Strictly equivalent
2	SDR relaxation (drop rank-1)	NC4	Tight relaxation (equiv. for $K \leq 2$)
3	S-Procedure closed-form bound	NC2	Conservative approx. (1.75 dB margin)
4	SINR fraction linearization	NC1 (comm.)	Strictly equivalent
5a	PCRB affine expansion	NC6	Strictly equivalent (Assumption 1)
5b	Sensing SINR linearization	NC1 (sens.)	Strictly equivalent
6	Power constraint (already convex)	—	Already convex
7	AP selection two-step decomp.	NC3	Heuristic lower bound

Key conclusion: The physical-layer convexification (Steps 1–6) is almost entirely strictly equivalent. Only Step 3 is a conservative engineering approximation, and Step 2 is tight for $K \leq 2$. AP selection (Step 7) uses a heuristic to maintain polynomial complexity.

5.0.4 Final Convex SDP Problem (P3)

After the seven-step chain, the original problem (P1) reduces to the following **standard convex SDP**:

$$
\begin{aligned}
\min_{\{\mathbf{W}_k\},\mathbf{Z},\{\mu_k\}} \quad & \sum_{k=1}^{K} \mathrm{tr}(\mathbf{W}_k) + \mathrm{tr}(\mathbf{Z}) && \text{(P3)} \\
\text{s.t.} \quad & \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k && (23) \\
& \mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2, \quad \forall p && (24) \\
& \mathrm{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p && (25) \\
& \mathrm{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m && (26) \\
& \mathbf{W}_k \succeq \mathbf{0}, \quad \forall k && (27) \\
& \mathbf{Z} \succeq \mathbf{0} && (28) \\
& \mu_k \geq 0, \quad \forall k && (29)
\end{aligned}
$$

where $\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j\neq k}\mathbf{W}_j$, $\mathbf{F}_p$ is given by Eq. (13), $\mathbf{E}_m$ is the selection matrix for AP m, and $\mu_k \geq 0$ is the S-Procedure slack variable.

Table 4: Problem (P3) constraint structure and variable mapping

Constraint	Type	Variables	Physical meaning
(31)	LMI ($MN_t + 1$ dim.)	$\mathbf{W}_k, \mu_k$	Communication worst-case robustness
(32)	Linear inequality	$\mathbf{Z}$	Sensing detection probability
(33)	Linear inequality	$\mathbf{R}_X$	Tracking accuracy (PCRB)
(34)	Linear inequality	$\mathbf{R}_X$	Per-AP peak power
(35)–(36)	PSD cone	$\mathbf{W}_k, \mathbf{Z}$	SDR relaxation
(37)	Non-negative orthant	μ_k	S-Procedure slack

Theorem 1 (Convexity of the problem)**.** *Problem (P3) is a standard convex SDP: the objective is linear and all constraints are linear matrix inequalities (LMIs) or semidefinite cone constraints. It can be solved to arbitrary accuracy in polynomial time by standard SDP solvers (MOSEK / SeDuMi / SDPT3). The solution complexity is* $O((MN_t)^{3.5})$*, taking approximately 5–10 seconds for* $M = 16, N_t = 4$.

Theorem 2 (SDR tightness)**.** *For the total power minimization problem, SDR is tight when* $K \leq 2$*, i.e., the optimal solution satisfies* $\mathrm{rank}(\mathbf{W}_k^*) = 1$*. In the high-SNR regime, SDR is approximately tight. For general* $K > 2$*, beam recovery via Gaussian randomization incurs a performance loss upper bounded by* $O(1/L)$ *(where* L *is the number of candidates).*

5.0.5 Complexity Comparison: Before and After Convexification

Convexification transforms an NP-hard mixed-integer non-convex problem into a polynomial-time-solvable standard convex SDP, which is the central value of the convexification methodology.

Table 5: Comparison of problem form and solution complexity

Problem	Form	Solver	Complexity
(P1) Original	MINLP (mixed-integer non-convex)	No general algorithm	NP-hard
(P3) Convex SDP	Convex semidefinite program	MOSEK / SeDuMi	$O((MN_t)^{3.5})$

5.1 From Beams to Covariances

To transform the non-convex beamforming problem into a convex optimization problem, we introduce covariance matrix variables:

Communication covariance matrix:

$$\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H \in \mathbb{C}^{MN_t \times MN_t} \tag{15}$$

Sensing covariance matrix:

$$\mathbf{Z} = \sum_p \mathbf{z}_p \mathbf{z}_p^H \in \mathbb{C}^{MN_t \times MN_t} \tag{16}$$

Global transmit covariance:

$$\mathbf{R}_X = \sum_{k=1}^{K} \mathbf{W}_k + \mathbf{Z} \tag{17}$$

5.2 SDR Relaxation and Final Convex SDP Problem

The original problem requires $\text{rank}(\mathbf{W}_k) = 1$ (since $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$). Semidefinite relaxation (SDR) discards the rank-one constraint, requiring only $\mathbf{W}_k \succeq \mathbf{0}$. Furthermore, through the S-Procedure, the infinite-dimensional worst-case SINR constraint is exactly transformed into a finite-dimensional linear matrix inequality (LMI), yielding the following **standard convex semidefinite programming problem** that can be directly fed into CVX/MOSEK.

Core Auxiliary Definition:

$$\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j \tag{30}$$

Final Convex SDP Problem (P3):

$$\min_{\{\mathbf{W}_k\}, \mathbf{Z}, \{\mu_k\}} \quad \sum_{k=1}^{K} \text{tr}(\mathbf{W}_k) + \text{tr}(\mathbf{Z}) \tag{P3}$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \tag{31}$$

$$\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \tag{32}$$

$$\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track},p}, \quad \forall p \tag{33}$$

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \tag{34}$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \tag{35}$$

$$\mathbf{Z} \succeq \mathbf{0} \tag{36}$$

$$\mu_k \geq 0, \quad \forall k \tag{37}$$

where $\mathbf{E}_m$ is the selection matrix for AP m, and $\mu_k \geq 0$ is the S-Procedure slack variable.

Table 6: Problem (P3) Constraint Structure and Variable Mapping

Constraint	Type	Variables	Physical Meaning
(31)	LMI ($MN_t + 1$ dim.)	$\mathbf{W}_k, \mu_k$	Communication worst-case robustness
(32)	Linear inequality	$\mathbf{Z}$	Sensing detection probability
(33)	Linear inequality	$\mathbf{R}_X$	Tracking accuracy (PCRB)
(34)	Linear inequality	$\mathbf{R}_X$	Per-AP peak power
(35)–(36)	PSD cone	$\mathbf{W}_k, \mathbf{Z}$	SDR relaxation
(37)	Non-negative orthant	μ_k	S-Procedure slack

5.3 Tightness Conditions

Theorem 3 (SDR Tightness)**.** *For the total power minimization problem, when* $K \leq 2$*, SDR is tight, i.e., the optimal solution satisfies* $\mathrm{rank}(\mathbf{W}_k^*) = 1$*.*

Remark 4. *In the high-SNR regime, SDR is approximately tight with controllable performance loss. For the general case, beam recovery via Gaussian randomization provides a performance loss upper bounded by* $O(1/L)$ *(where* L *is the number of candidates).*

6 S-Procedure Robust Constraint Transformation

6.1 Communication Robust Constraint (Exact Transformation)

The worst-case SINR constraint is:

$$\min_{\|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h} \frac{|(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_k|^2}{\sum_{j \neq k} |(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_j|^2 + \sigma_c^2} \geq \gamma_k \tag{38}$$

Using the S-Procedure, this infinite-dimensional constraint can be transformed into a finite-dimensional LMI. Define:

$$\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j \tag{39}$$

Then there exists $\mu_k \geq 0$ such that the following LMI holds:

$$\begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0} \tag{18}$$

Proposition 6. *The S-Procedure is an* ***exactly equivalent*** *transformation, not an approximation. That is, the original worst-case constraint and the LMI constraint are completely equivalent.*

6.2 Sensing Robust Constraint (Optional)

Similarly, for the sensing channel, the LMI form is:

$$\begin{bmatrix} \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z} + \nu_p \mathbf{I} & \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z}\hat{\mathbf{g}}_p \\ \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z} & \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z}\hat{\mathbf{g}}_p - \sigma_s^2 - \nu_p \epsilon_g^2 \end{bmatrix} \succeq \mathbf{0} \tag{19}$$

where $\nu_p \geq 0$ is the sensing S-Procedure slack variable.

Remark 5 (LMI Typesetting Optimization). *The matrix elements in (19) contain denominators γ_S^{PoD}. In standard LMI notation, both sides of the inequality can be multiplied by γ_S^{PoD}, yielding $\mathbf{Z} + \nu_p \gamma_S^{PoD} \mathbf{I}$ in the upper-left corner, thereby avoiding fractional structures inside the matrix for more elegant typesetting. Mathematical equivalence is preserved.*

Assumption 2 (Sensing Channel Perfection). *This work focuses on robust design for the communication link, assuming that sensing channels are self-calibrated through tracking echoes within the current time slot with negligible error. Specifically:*

1. *Target state parameters $\boldsymbol{\theta}_p$ are accurately predicted within a short time slot, with prediction errors much smaller than one wavelength*
2. *Sensing channels $\mathbf{g}_p$ are calibrated through tracking echoes from the previous time slot, with errors incorporated into the next slot update*
3. *Sensing tasks adopt a "detection-tracking" cascade architecture: conservative thresholds during detection, and time-slot smoothness compensates for errors during tracking*

Remark 6. *Sensing channels are self-calibrating (the probing signal is transmitted and the echo is received, with the echo carrying current channel state), whereas communication channels rely on uplink pilot estimation, where pilot contamination leads to error accumulation. In top-tier journal system modeling, focusing on communication-side pilot contamination while assuming the radar side benefits from LOS echoes and Kalman tracking filtering for precise prediction is a common and prudent compromise.*

7 Proof of Sensing Constraint Convexity

This section proves the convexity of the sensing-side constraints that were intentionally left in their non-convex form during the seven-step chain.

7.1 PCRB Constraint Convexity

Constraint: $\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track},p}$

- $\mathbf{F}_p$ is a known constant positive semidefinite matrix
- $\mathbf{R}_X$ is the optimization variable (positive semidefinite matrix)
- $\text{tr}(\mathbf{F}_p \mathbf{R}_X)$ is a linear function of $\mathbf{R}_X$

Conclusion: Linear inequality constraint, naturally convex.

7.2 Sensing SINR Constraint Convexity

Constraint: $\mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}} \sigma_s^2$

- $\mathbf{g}_p \mathbf{g}_p^H$ is a known constant positive semidefinite matrix
- $\mathbf{Z}$ is the optimization variable
- $\mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z})$ is a linear function of $\mathbf{Z}$

Conclusion: Linear inequality constraint, naturally convex.

7.3 Power Constraint Convexity

Constraint: $\mathrm{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}$

- $\mathbf{E}_m$ is a known constant matrix
- $\mathbf{R}_X$ is the optimization variable

Conclusion: Linear inequality constraint, naturally convex.

7.4 Summary

Table 7 summarizes the convexity of each constraint.

Table 7: Convexity Summary of Constraints

Constraint	Form	Convexity
Communication Robust (S-Procedure)	LMI	Convex
PCRB	Linear trace	Convex
Sensing SINR	Linear trace	Convex
Power	Linear trace	Convex
Positive Semidefinite	$\mathbf{W}_k, \mathbf{Z} \succeq \mathbf{0}$	Convex

Theorem 4 (Final Problem Convexity)**.** *Problem (P3) is a standard convex SDP, solvable to arbitrary precision via interior-point methods in polynomial time. Combined with Algorithm 22 (Gaussian randomization rank-one recovery), this theoretical framework can be directly translated into MATLAB/CVX code.*

8 Rank-One Recovery and Fallback Scheme (Plate III: Engineering Feasibility)

The solution of (P3) is a covariance matrix; to obtain the physically realizable rank-one beamforming vector, an additional recovery step is required. This section presents Gaussian randomization together with a power-scaling fallback to convert the convex optimum of (P3) into a deployable beamforming vector.

8.1 Problem Background

SDR relaxation discards the $\text{rank}(\mathbf{W}_k) = 1$ constraint. After solving:

- If $\text{rank}(\mathbf{W}_k^*) = 1$: directly recover $\mathbf{w}_k$ through eigenvalue decomposition
- If $\text{rank}(\mathbf{W}_k^*) > 1$: Gaussian randomization is required to extract a suboptimal beam

8.2 Algorithm 1: Gaussian Randomization Rank-One Recovery

The following pseudocode describes the rank-one recovery algorithm. The algorithm first checks the rank of the communication covariance matrix; if the rank is 1, the beam is directly extracted. If the rank exceeds 1, candidate beams are generated via Gaussian randomization, and the best candidate satisfying the constraints is selected. If still unsatisfied, power scaling is employed as a fallback.

8.3 Power Scaling Mathematical Guarantee

After power scaling:

- Per-AP power: $P'_m = \beta_m P_m = P_{\max}$ (strictly satisfied)
- Communication SINR: $\text{SINR}'_k = \beta_m \cdot \text{SINR}_k$ (linear scaling)
- Sensing SINR: $\text{SINR}'_{S,p} = \beta_m \cdot \text{SINR}_{S,p}$ (linear scaling)

Remark 7. *The rank of the sensing covariance* $\mathbf{Z}^*$ *exceeding 1 is a* ***design intention*** *(multi-target coverage), not a recovery failure. In pure communication MU-MIMO, power minimization tends to form "pencil beams" (rank-one); in ISAC, sensing tasks require spatial energy dispersion, and multi-rank is a design intention. The "role separation" between communication and sensing in the covariance domain is the essential characteristic of ISAC waveform design.*

9 Closed-Form Solution Derivation (Plate IV: Benchmark)

As a benchmark against (P3) in Section III, this section derives a suboptimal closed-form solution based on ZF and matched filtering. This solution does not guarantee strict satisfaction of the robust constraints (success rate approximately 25%), but its complexity is only $O(K^3)$, making it an indispensable engineering baseline.

9.1 Assumptions

To obtain closed-form solutions, the problem is simplified under specific conditions:

- Ignore PCRB constraints (or assume they are automatically satisfied)
- Use ZF communication beams
- Use matched-filter sensing beams
- Assume relaxed per-AP power constraints

Algorithm 1: Gaussian Randomization Rank-One Recovery

Input: SDP optimal solution $\{\mathbf{W}_k^*\}_{k=1}^K$, $\mathbf{Z}^*$, constraint parameters
Output: Beams $\{\mathbf{w}_k\}_{k=1}^K$ satisfying all constraints, sensing waveform

`// Step 1: Communication beam recovery`

foreach *User* k **do**
  if $\mathrm{rank}(\mathbf{W}_k^*) = 1$ **then**
    $\mathbf{w}_k \leftarrow \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \cdot \mathbf{v}_{\max}(\mathbf{W}_k^*)$; `// Direct extraction`
  else
    Generate $L = 1000$ candidates: $\boldsymbol{\xi}_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$;
    Normalize: $\mathbf{w}_k^{(l)} \leftarrow \sqrt{\mathrm{tr}(\mathbf{W}_k^*)} \cdot \frac{\boldsymbol{\xi}_l}{\|\boldsymbol{\xi}_l\|_2}$;
    Compute constraint violation:
    $v_l \leftarrow \sum_{k'} \max(0, \gamma_k - \mathrm{SINR}_{k'}^{(l)}) + \sum_p \max(0, \gamma_S^{\mathrm{PoD}} - \mathrm{SINR}_{S,p}^{(l)})$;
    $l^* \leftarrow \arg\min_l v_l$;
    if $v_{l^*} = 0$ **then**
      Accept $\mathbf{w}_k \leftarrow \mathbf{w}_k^{(l^*)}$;
    else
      Enter power scaling fallback (Step 3);

`// Step 2: Sensing waveform recovery`

Eigenvalue decomposition: $\mathbf{Z}^* = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^H$;
Generate multi-stream transmission waveform: $\mathbf{z}_p = \sum_{i=1}^r \sqrt{\lambda_i} \mathbf{v}_i s_i$, where $s_i \sim \mathcal{CN}(0, 1)$ are independent;

`// Step 3: Power scaling fallback`

foreach *AP* m **do**
  $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m)$;
  if $P_m > P_{\max}$ **then**
    $\beta_m \leftarrow P_{\max}/P_m$;
    $\mathbf{w}_{m,k} \leftarrow \mathbf{w}_{m,k} \cdot \sqrt{\beta_m}$;
    $\mathbf{Z}_m \leftarrow \mathbf{Z}_m \cdot \beta_m$;

`// Step 4: Performance guarantee`

if $K \le 2$ **then** SDR is tight, optimal solution recovered;
if $K > 2$ **then** Gaussian randomization provides suboptimal solution with performance loss upper bounded by $O(1/L)$;

9.2 ZF Beamforming

Condition: $\text{rank}(\mathbf{H}_{\text{all}}) \geq K$, i.e., $M^{\text{all}} N_t \geq K$.

The ZF matrix is:

$$\mathbf{W}_{\text{ZF}} = \mathbf{H}_{\text{all}} (\mathbf{H}_{\text{all}}^H \mathbf{H}_{\text{all}})^{-1} \tag{20}$$

Satisfying the orthogonality property:

$$\mathbf{h}_k^{\text{all},H} \mathbf{W}_{\text{ZF}}(:, j) = \delta_{kj} \tag{40}$$

Power allocation:

$$p_k = \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:, k)\|_2^2 \tag{21}$$

9.3 Sensing Beams

$$\mathbf{z}_p = \sqrt{P_{S,p}} \frac{\mathbf{g}_p^{\text{all}}}{\|\mathbf{g}_p^{\text{all}}\|_2} \tag{22}$$

where $P_{S,p} = (\gamma_S^{\text{PoD}})^{\text{robust}} \sigma_s^2 / \|\mathbf{g}_p^{\text{all}}\|_2^2$.

9.4 Algorithm 2: Closed-Form Solver

The following pseudocode describes the closed-form solver. The algorithm first selects the optimal AP subset for each target based on sensing channel strength, then computes ZF communication beams and matched-filter sensing beams on the selected APs, and finally verifies power constraints before returning a feasible solution.

10 Complexity, Tightness, and Feasibility Analysis

As an extension of Section III, this section presents the SDR tightness theorem (tight for $K \leq 2$, approximately tight in the high-SNR regime), the necessary and sufficient feasibility conditions, and a complexity comparison between (P3) and the closed-form solution.

10.1 SDP Solving Complexity

Number of variables:

- K matrices $\mathbf{W}_k$: $K \cdot (MN_t)^2$ real variables
- 1 matrix $\mathbf{Z}$: $(MN_t)^2$ real variables
- K variables μ_k: K real variables
- Total: $O(K(MN_t)^2)$

Number of constraints:

- K communication LMIs: $K \cdot (MN_t + 1) \times (MN_t + 1)$
- P sensing linear constraints
- M power constraints

Algorithm 2: Cell-Free ISAC Closed-Form Solver

Input: $\mathbf{H}, \mathbf{G}, M, N_t, K, P, P_{\max}, \{\gamma_k\}, \gamma_S^{\text{PoD}}, N_{\text{req}}, \epsilon_h, \epsilon_g$
Output: $\{\mathbf{w}_{m,k}\}, \{\mathbf{Z}_m\}, \{b_{mp}\}, P_{\text{total}}$
`// Step 1: Compute robust thresholds`
$\gamma_k^{\text{robust}} \leftarrow \gamma_k \cdot (1+\epsilon_h)^2/(1-\epsilon_h)^2$;
$\gamma_S^{\text{robust}} \leftarrow \gamma_S^{\text{PoD}} \cdot (1+\epsilon_g)^2/(1-\epsilon_g)^2$;
`// Step 2: AP selection`
for $p = 1, \ldots, P$ **do**
  Compute $\|\mathbf{g}_{m,p}\|_2$ for all m;
  Select top-N_{req} APs: $b_{mp} = 1$;
$\mathcal{M}^{\text{all}} \leftarrow \bigcup_p \{m : b_{mp} = 1\}$;
`// Step 3: Extract sub-channels`
$\mathbf{H}^{\text{all}} \leftarrow \mathbf{H}(\mathcal{M}^{\text{all}}, :)$;
$\mathbf{g}_p^{\text{all}} \leftarrow \mathbf{G}(\mathcal{M}^{\text{all}}, p)$ for all p;
`// Step 4: Communication beams`
if $\text{rank}(\mathbf{H}^{all}) \geq K$ **then**
  $\mathbf{W}_{\text{ZF}} \leftarrow \mathbf{H}^{\text{all}} \cdot (\mathbf{H}^{\text{all},H}\mathbf{H}^{\text{all}})^{-1}$;
  for $k = 1, \ldots, K$ **do**
    $\mathbf{w}_k^{\text{ZF}} \leftarrow \mathbf{W}_{\text{ZF}}(:, k)/\|\mathbf{W}_{\text{ZF}}(:, k)\|_2$;
    $p_k \leftarrow \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:, k)\|_2^2$;
    $\mathbf{w}_k \leftarrow \sqrt{p_k} \cdot \mathbf{w}_k^{\text{ZF}}$;
else
  Use MRT (suboptimal);
`// Step 5: Sensing beams`
for $p = 1, \ldots, P$ **do**
  $P_{S,p} \leftarrow \gamma_S^{\text{robust}} \sigma_s^2 / \|\mathbf{g}_p^{\text{all}}\|_2^2$;
  $\mathbf{z}_p \leftarrow \sqrt{P_{S,p}} \cdot \mathbf{g}_p^{\text{all}} / \|\mathbf{g}_p^{\text{all}}\|_2$;
  $\mathbf{Z}_m \leftarrow \mathbf{z}_p \mathbf{z}_p^H$ (rank-1);
`// Step 6: Power check`
for $m \in \mathcal{M}^{all}$ **do**
  $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m)$;
  if $P_m > P_{\max}$ **then**
    Scale or mark infeasible;
`// Step 7: Verify and return`

- Total computational complexity: $O(K(MN_t)^3)$

Solving time: $O((MN_t)^{3.5})$ — for $M = 16, N_t = 4$, approximately 5–10 seconds (MOSEK).

10.2 Closed-Form Solution Complexity

Table 8 lists the complexity of each step in the closed-form solver.

Table 8: Complexity Analysis of Closed-Form Solver

Step	Operation	Complexity
AP Selection	Sorting per target	$O(MP \log M)$
ZF Inversion	$(\mathbf{H}^H\mathbf{H})^{-1}$	$O(K^3)$
Communication Power	K multiplications	$O(K)$
Sensing Beams	P normalizations	$O(PMN_t)$
Power Check	M summations	$O(MK)$
Total		$O(MP \log M + K^3)$

For $M = 16, N_t = 4, K = 10, P = 4$: $O(7400)$, i.e., $less than 0.1$ second.

11 Feasibility Conditions

11.1 Necessary Conditions

ZF feasibility:

$$M^{\mathrm{all}} N_t \geq K \tag{41}$$

For $N_t = 4, K = 10$: $M^{\mathrm{all}} \geq 3$ (theoretical), $M^{\mathrm{all}} \geq 4$ (numerically stable).
Power feasibility:

$$P_{\mathrm{comm}}^{\min} + P_{\mathrm{sens}}^{\min} \leq M \cdot P_{\max} \tag{42}$$

where:

$$P_{\mathrm{comm}}^{\min} = \sum_{k=1}^{K} \gamma_k^{\mathrm{robust}} \sigma_c^2 \|\mathbf{W}_{\mathrm{ZF}}(:,k)\|_2^2 \tag{43}$$

$$P_{\mathrm{sens}}^{\min} = \sum_{p=1}^{P} (\gamma_S^{\mathrm{PoD}})^{\mathrm{robust}} \frac{\sigma_s^2}{\|\mathbf{g}_p^{\mathrm{all}}\|_2^2} \tag{44}$$

11.2 Infeasible Scenarios

1. Target is far from all APs ($d > 50$ m)

2. Excessive number of users ($K > M^{\mathrm{all}} N_t$)

3. Insufficient power budget ($P_{\max} < P_{\mathrm{comm}}^{\min}$)

4. Excessive CSI error ($\epsilon > 0.5$)

12 Key Formulas Summary

For ease of reference, this section consolidates the core formulas of the paper.

Table 9 summarizes the core formulas of this paper.

Table 9: Summary of Key Formulas

Number	Name	Expression
(3)	Communication robustness factor	$\eta_h = \left(\frac{1-\epsilon_h}{1+\epsilon_h}\right)^2$
(4)	Communication robust threshold	$\gamma_k^{\text{robust}} = \gamma_k/\eta_h$
(9)	Sensing robustness factor	$\eta_g = \left(\frac{1-\epsilon_g}{1+\epsilon_g}\right)^2$
(10)	Sensing robust threshold	$(\gamma_S^{\text{PoD}})^{\text{robust}} = \gamma_S^{\text{PoD}}/\eta_g$
(12)	FIM trace constraint	$\text{tr}(\mathbf{J}_p^{\text{data}}) = \text{tr}(\mathbf{F}_p\mathbf{R}_X)$
(13)	FIM constant matrix	$\mathbf{F}_p = \frac{2}{\sigma_s^2}\text{Re}\left\{\nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p \nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p^H\right\}$
(18)	Communication S-Procedure LMI	$\begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}$
(19)	Sensing S-Procedure LMI (optional)	$\begin{bmatrix} \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z} + \nu_p \mathbf{I} & \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z}\hat{\mathbf{g}}_p \\ \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z} & \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z}\hat{\mathbf{g}}_p - \sigma_s^2 - \nu_p \epsilon_g^2 \end{bmatrix} \succeq \mathbf{0}$
(20)	ZF matrix	$\mathbf{W}_{\text{ZF}} = \mathbf{H}(\mathbf{H}^H\mathbf{H})^{-1}$
(21)	Communication power allocation	$p_k = \gamma_k^{\text{robust}}\sigma_c^2\|\mathbf{W}_{\text{ZF}}(:,k)\|_2^2$

13 Conclusion and Future Work

This work establishes a complete theoretical framework for robust beamforming in cell-free ISAC MIMO systems, spanning physical modeling, mathematical reformulation, engineering realization, and benchmark baseline. The core contributions are fourfold. First, the highly non-convex MINLP (P1) is transformed, via a seven-step rigorous derivation, into a standard convex SDP (P3). Second, each step of the transformation is accompanied by an explicit identification of the non-convex source, the pre- and post-transformation form, and a strict equivalence, tightness, or conservative-boundedness proof. Third, a Gaussian-randomization and power-scaling fallback algorithm is provided so that the convex optimum of (P3) can be converted into a physically realizable rank-one beamforming vector. Fourth, a closed-form ZF-plus-matched-filtering baseline is proposed as a comparison reference.

Future work includes: implementing a full MOSEK-based SDP solver and verifying the tightness boundary of (P3); investigating multi-target time-slot coupled MPC; extending the present framework to RIS-aided ISAC scenarios; and benchmarking against recent deep reinforcement learning approaches (such as the cell-free ISAC beam selection work by Derrick Wing Kwan Ng and colleagues in 2024) on a unified platform.

This paper establishes a complete mathematical framework for joint beamforming optimization in Cell-Free ISAC systems. The main contributions include:

1. **Problem Formulation**: Established the standard optimization problem (P1) incorporating communication QoS, sensing detection, tracking accuracy, and AP se-

lection constraints.

2. **Robustness Analysis**: Derived worst-case SINR constraints under CSI errors, introducing robustness factors η_h and η_g to quantify the impact of estimation errors.

3. **Convex Relaxation**: Transformed the non-convex rank-one constraint into a semidefinite constraint via SDR, and proved the convexity of sensing constraints under the single-angle estimation assumption.

4. **S-Procedure Transformation**: Exactly transformed infinite-dimensional worst-case constraints into finite-dimensional LMIs, guaranteeing mathematical equivalence.

5. **Rank-One Recovery**: Proposed Gaussian randomization algorithms with power scaling fallback schemes to ensure solution feasibility.

6. **Closed-Form Solution**: Provided ZF beams with matched-filter sensing under simplified conditions, with complexity $O(MP \log M + K^3)$.

Future work:

- Implement the SDP solver (requires MOSEK support for LMI constraints)
- Extend sensing channel robustness (Options B/C)
- Multi-target joint optimization verification
- Simulation validation and performance evaluation